

EXERCISE SHEET (IV)

MULTIPLE INTEGRALS

(1) Calculate the following iterated integral

(a) $\int_1^3 \int_0^1 (1+4xy) dx dy$

(b) $\int_1^4 \int_1^2 \left(\frac{x}{y} + \frac{y}{x} \right) dy dx$

(c) $\int_0^2 \int_0^1 (2x+xy)^8 dx dy$

(d) $\int_0^{\ln 2} \int_0^{\ln 5} e^{2x-y} dx dy$

(e) $\int_0^1 \int_0^1 \frac{xy}{\sqrt{x^2+y^2+1}} dy dx$

(e) $\int_0^\pi \int_0^{\pi/2} \cos(x+2y) dy dx$

(f) $\int_0^1 \int_{-3}^3 \frac{xy^2}{x^2+1} dy dx$

(g) $\int_0^{\pi/6} \int_0^{\pi/3} x \sin(x+y) dy dx$

(h) $\int_0^2 \int_0^1 xy e^{x^2 y} dx dy$

(i) $\int_0^1 \int_1^2 \frac{x}{x^2+y^2} dx dy$

(a) $\int_1^3 \int_0^1 (1+4xy) dx dy = \int_1^3 [x + 2x^2 y]_0^1 dy = \int_1^3 (1+2y) dy = y + y^2 \Big|_1^3 = 10$

(b) $\int_1^4 \int_1^2 \left(\frac{x}{y} + \frac{y}{x} \right) dy dx = \int_1^4 \left[x \cdot \ln|y| + \frac{1}{x} \cdot \frac{y^2}{2} \right]_{y=1}^{y=2} dx =$

$\int_1^4 \left[\left(x \ln 2 + \frac{2}{2x} \right) - \left(x \ln 1 + \frac{1}{2x} \right) \right] dx = \int_1^4 \left(x \ln 2 + \frac{3}{2x} \right) dx$

$= \frac{x^2}{2} \ln 2 + \frac{3}{2} \ln|x| \Big|_1^4 = \frac{16}{2} \ln 2 + \frac{3}{2} \ln 4 - \frac{\ln 2}{2} - \frac{\ln 1 \cdot 3}{2}$

$= \frac{15}{2} \ln 2 + 3 \ln 4^{1/2}$

$$\textcircled{c} \int_0^2 \int_0^1 (2x+y)^8 dx dy = \int_0^2 \frac{1}{2} \frac{(2x+y)^9}{9} \Big|_0^1 dy = \frac{1}{18} \int_0^2 [(2+y)^9 - y^9] dy$$

$$= \frac{1}{18} \left[\frac{(2+y)^{10}}{10} - \frac{y^{10}}{10} \right]_0^2 = \frac{1}{180} (4^{10} - 2^{10})$$

$$\textcircled{d} \int_0^{\ln 2} \int_0^{\ln 5} e^{2x-y} dx dy = \int_0^{\ln 2} \frac{e^{2x-y}}{2} \Big|_0^{\ln 5} dy = \int_0^{\ln 2} \left(\frac{e}{2} - \frac{1}{2} \right) e^{-y} dy$$

$$= - \left(\frac{e}{2} - \frac{1}{2} \right) e^{-y} \Big|_0^{\ln 2} = - \left(\frac{25}{2} - \frac{1}{2} \right) \cdot [e^{-\ln 2} - 1]$$

$$(R_{\text{ule}} e^{a \cdot \ln b} = e^{\ln b^a} = b^a) \quad = -12 \cdot \left[\frac{1}{2} - 1 \right] = \boxed{6}$$

$$\textcircled{e} \int_0^{\pi} \int_0^{\pi/2} \cos(x+2y) dy dx = \int_0^{\pi} \frac{1}{2} \sin(x+2y) \Big|_0^{\pi/2} dx = \frac{1}{2} \int_0^{\pi} [\sin(x+\pi) - \sin x] dx$$

$$= \frac{1}{2} [-\cos(x+\pi) + \cos x] \Big|_0^{\pi} = \frac{1}{2} [-\cos(2\pi) + \cos \pi - (\cos \pi + \cos 0)]$$

$$= \frac{1}{2} (-1 - 1 - (1 + 1)) = -2$$

$$\textcircled{e} \int_0^1 \int_0^1 \frac{xy}{\sqrt{x^2+y^2+1}} dy dx = \int_0^1 x \sqrt{x^2+y^2+1} \Big|_0^1 dx = \int_0^1 x \sqrt{x^2+2} - x \sqrt{x^2+1} dx$$

$$= \frac{1}{3} [(x^2+2)^{3/2} - (x^2+1)^{3/2}] \Big|_0^1$$

$$= \frac{1}{3} [3^{3/2} - 2^{3/2} - (2^{3/2} - 1)]$$

$$= \frac{1}{3} (3\sqrt{3} - 4\sqrt{2} + 1)$$

$\sqrt{x^2+y^2+1} = u$
 $\frac{\partial}{\partial y} \sqrt{x^2+y^2+1} = \frac{y}{\sqrt{x^2+y^2+1}} = \frac{1}{u}$
 $\frac{\partial}{\partial y} \frac{1}{u} = -\frac{1}{u^2} = -\frac{1}{(x^2+y^2+1)^{3/2}}$

$$\textcircled{f} \int_0^1 \int_{-3}^3 \frac{xy^2}{x^2+1} dy dx = \int_0^1 \frac{x}{x^2+1} \cdot \frac{y^3}{3} \Big|_{-3}^3 dx = \frac{\pi 4}{3} \int_0^1 \frac{x}{x^2+1} dx$$

$$x^2+1=u \quad 2x dx = du$$

$$= \frac{\pi 4}{3} \frac{1}{2} \ln(x^2+1) \Big|_0^1$$

$$= \frac{2\pi}{3} \ln 2$$

$$\textcircled{g} \int_0^{\pi/6} \int_0^{\pi/3} x \sin(x+y) dy dx = \int_0^{\pi/6} -x \cos(x+y) \Big|_{y=0}^{y=\pi/3} dx$$

$$= \int_0^{\pi/6} [-x \cos(x+\pi/3) + x \cos(x)] dx = x [\sin x - \sin(x+\pi/3)] \Big|_0^{\pi/6}$$

$$+ [\cos x - \cos(x+\pi/3)] \Big|_0^{\pi/6} = \frac{\sqrt{3}-1}{2} - \frac{\pi}{12}$$

$$\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x$$

$$x=u \quad \cos x dx = du$$

$$dx = du \quad \sin x = u$$

$$\int x \cos(x+\pi/3) dx = x \sin(x+\pi/3) - \int \sin(x+\pi/3)$$

$$x=u \quad \cos(x+\pi/3) dx = du$$

$$= x \sin(x+\pi/3) + \cos(x+\pi/3)$$

$$dx = du \quad \sin(x+\pi/3) = u$$

$$\textcircled{h} \int_0^2 \int_0^1 xy e^{x^2 y} dx dy = \int_0^2 \left[\frac{1}{2} e^{x^2 y} \right]_{x=0}^{x=1} dy = \frac{1}{2} \int_0^2 (e^y - 1) dy$$

$$= \frac{1}{2} [e^y - y]_0^2 = \frac{1}{2} [(e^2 - 2) - (1 - 0)] = \frac{1}{2} (e^2 - 3)$$

$$\textcircled{i} \int_0^1 \int_1^2 \frac{x}{x^2+y^2} dx dy = \int_0^1 \left[\frac{1}{2} \ln(x^2+y^2) \right]_{x=1}^2 dy = \frac{1}{2} \int_0^1 [\ln(4+y^2) - \ln(1+y^2)] dy$$

$$\int \ln(4+y^2) dy = \ln(4+y^2) \cdot y - \int \frac{2y}{4+y^2} dy = \ln(4+y^2) y - \int \left(2 - \frac{8}{4+y^2} \right) dy$$

$$\ln(4+y^2) = u \quad \frac{2y}{4+y^2} = du$$

$$dy = dv$$

$$y = v$$

$$= y \ln(4+y^2) - 2y + \frac{8}{2} \arctan\left(\frac{y}{2}\right)$$

$$\int \ln(1+y^2) dy = y \ln(1+y^2) - 2y + 2 \arctan y$$

(3)

Deyu Altıntaş

$$\int_0^1 \int_1^2 \frac{x}{x^2+y^2} dx dy = \frac{1}{2} [y \ln(4+y^2) - 2y + 4 \operatorname{arctan}(\frac{y}{2}) - y \ln(1+y^2)$$

$$+ 2y - 2 \operatorname{arctan} y] \Big|_0^1 = \frac{1}{2} [(\ln 5 + 4 \operatorname{arctan}(\frac{1}{2}) - \ln 2 - 2 \operatorname{arctan} 1) - 0]$$

$$= \frac{1}{2} \ln \frac{5}{2} + 2 \operatorname{arctan}(\frac{1}{2}) - \frac{\pi}{4}$$

② Find the volume of the solid that lies under the plane.

$3x + 2y + z = 12$ and above the rectangle

$$R = \{(x, y) \mid 0 \leq x \leq 1, -2 \leq y \leq 3\}$$

$$V = \int_{-2}^3 \int_0^1 (12 - 3x - 2y) dx dy = \int_{-2}^3 [12x - \frac{3}{2}x^2 - 2xy] \Big|_{x=0}^{x=1} dy$$

$$= \int_{-2}^3 (\frac{21}{2} - 2y) dy = [\frac{21}{2}y - y^2] \Big|_{-2}^3 = \frac{95}{2}$$

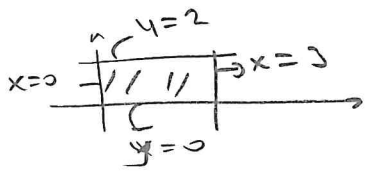
③ Find the volume of the solid lying under the elliptic paraboloid $\frac{x^2}{4} + \frac{y^2}{9} + z = 1$ and above the rectangle $R = [-1, 1] \times [-2, 2]$

$$V = \int_{-2}^2 \int_{-1}^1 [1 - \frac{x^2}{4} - \frac{y^2}{9}] dx dy = \int_{-2}^2 [x - \frac{x^3}{12} - \frac{xy^2}{9}] \Big|_{-1}^1 dy$$

$$= \int_{-2}^2 [(1 - \frac{1}{12} - \frac{y^2}{9}) - (-1 + \frac{1}{12} + \frac{y^2}{9})] dy = \int_{-2}^2 (\frac{22}{12} - \frac{2y^2}{9}) dy$$

$$= \frac{22}{12} y - \frac{2}{9} \cdot \frac{y^3}{3} \Big|_{-2}^2 = (\frac{44}{12} - \frac{2}{9} \cdot \frac{8}{3}) - (-\frac{44}{12} + \frac{16}{27}) = \frac{166}{27}$$

- 4) Find the volume of the solid bounded by the elliptic paraboloid $z = 1 + (x-1)^2 + 4y^2$ the planes $x=3$ and $y=2$ and the coordinate planes.



$$\int_0^3 \int_0^2 (1 + (x-1)^2 + 4y^2) dy dx = \int_0^3 \left[y + y(x-1)^2 + \frac{4y^3}{3} \right]_0^2 dx$$

$$= \int_0^3 \left(2 + 2(x-1)^2 + \frac{32}{3} \right) dx = \frac{38}{3}x + \frac{2(x-1)^3}{3} \Big|_0^3 = 44$$

- 5) Find the volume of the solid in the first octant bounded by the $z = 9 - y^2$ and the plane $x = 2$

Since it is in the first octant $z \geq 0$ so $y \leq 3$.

This means that we have $0 \leq y \leq 3$ $0 \leq x \leq 2$.

$$\int_0^3 \int_0^2 (9 - y^2) dx dy = \int_0^3 (9 - y^2) \cdot x \Big|_0^2 dy = \int_0^3 (18 - 2y^2) dy = 18y - \frac{2y^3}{3} \Big|_0^3 = 45$$

- 6) Evaluate the following integrals

a) $\int_0^1 \int_0^{x^2} (x + 2y) dy dx$

b) $\int_0^1 \int_x^{2-x} (x^2 - y) dy dx$

c) $\int_0^{\pi/2} \int_0^{\cos \theta} e^{r \cos \theta} dr d\theta$

d) $\int_0^1 \int_0^v \sqrt{1-v^2} dv dv$

e) $\iint_D x^3 y^2 dA$

$D = \{(x, y) \mid 0 \leq x \leq 2, -x \leq y \leq x\}$

f) $\iint_D \frac{2y}{x^2 + 1} dA$

$D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq \sqrt{x}\}$

g) $\iint_D e^{y^2} dA$

$D = \{(x, y) \mid 0 \leq y \leq 1, 0 \leq x \leq y\}$

h) $\iint_D x \cos y dA$

D is bounded by $y=0, y=x^2, x=1$

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Deyu Altintor

(I) $\iint_D xy^2 dA$ D is enclosed by $x=0$ $x=\sqrt{1-y^2}$

(Q) $\int_0^1 \int_0^{x^2} (x+2y) dy dx = \int_0^1 [xy + y^2]_{y=0}^{y=x^2} dx = \int_0^1 [x(x^2) + (x^2)^2 - 0] dx$
 $= \int_0^1 (x^3 - x^4) dx = \frac{x^4}{4} + \frac{x^5}{5} \Big|_0^1 = \frac{9}{20}$

(b) $\int_0^1 \int_x^{2-x} (x^2 - y) dy dx = \int_0^1 (x^2 y - \frac{y^2}{2}) \Big|_x^{2-x} dx =$

$\int_0^1 [x^2(2-x) - (\frac{2-x}{2})^2] - [x^3 - \frac{x^2}{2}] dx = [2\frac{x^3}{3} - \frac{x^4}{4} + \frac{(2-x)^3}{6}$
 $- \frac{x^4}{4} + \frac{x^3}{6}] \Big|_0^1 = \frac{5}{6}$

(c) $\int_0^{\pi/2} \int_0^{\cos \theta} e^{\sin \theta} r dr d\theta = \int_0^{\pi/2} e^{\sin \theta} \Big|_0^{\cos \theta} d\theta = \int_0^{\pi/2} e^{\sin \theta} \cdot \cos \theta d\theta$
 $= e^{\sin \theta} \Big|_0^{\pi/2} = e^{\sin(\pi/2)} - e^{\sin 0} = e - 1$

(d) $\int_0^1 \int_0^v \sqrt{1-v^2} du dv = \int_0^1 \sqrt{1-v^2} \underbrace{v}_{-\frac{du}{2}} dv = \frac{-1(1-v^2)^{3/2}}{3/2} \Big|_0^1 = \frac{1}{3}$

(e) $\int_0^2 \int_{-x}^x x^3 y^2 dy dx = \int_0^2 x^3 \cdot \frac{y^3}{3} \Big|_{-x}^x dx = \int_0^2 \frac{2x^6}{3} dx = \frac{2}{3} \frac{x^7}{7} \Big|_0^2 = \frac{256}{21}$

(f) $\int_0^1 \int_0^{\sqrt{x}} \frac{2y}{x^2+1} dy dx = \int_0^1 \left[\frac{y^2}{x^2+1} \right]_{y=0}^{y=\sqrt{x}} dx = \int_0^1 \frac{x}{x^2+1} dx$
 $= \frac{1}{2} \ln|x^2+1| \Big|_0^1 = \frac{1}{2} \ln 2 - \ln 1 = \frac{1}{2} \ln 2$

$$\textcircled{g} \int_0^1 \int_0^y e^{y^2} dx dy = \int_0^1 (e^{y^2} \cdot x) \Big|_0^y dy = \int_0^1 e^{y^2} \cdot y \underbrace{dy}_{\frac{du}{2}} = \frac{e^{y^2}}{2} \Big|_0^1 = \frac{e-1}{2}$$

$$\textcircled{h} \int_0^1 \int_0^{x^2} x \cos y dy dx = \int_0^1 x \sin y \Big|_0^{x^2} dx = \int_0^1 (x \sin x^2 - x \cdot \sin 0) dx$$

$x^2 = u \quad 2x dx = du$

$$= -\frac{\cos(x^2)}{2} \Big|_0^1 = \frac{(1 - \cos 1)}{2}$$

$$\textcircled{i} \int_{-1}^1 \int_0^{\sqrt{1-y^2}} xy^2 dx dy = \int_{-1}^1 y^2 \cdot \frac{1}{2} x^2 \Big|_0^{\sqrt{1-y^2}} dy = \frac{1}{2} \int_{-1}^1 y^2 (1-y^2) dy$$

$$= \frac{1}{2} \int_{-1}^1 (y^2 - y^4) dy = \frac{1}{2} \left(\frac{y^3}{3} - \frac{y^5}{5} \right) \Big|_{-1}^1 = \frac{2}{15}$$

$\textcircled{7}$ Find the volume of the given solid.

(a) Under the plane $x+2y-z=0$ and above the region bounded by $y=x$ and $y=x^4$

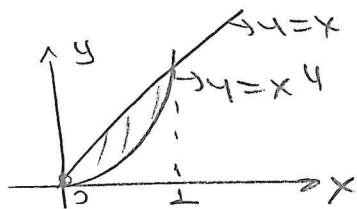
(b) Under the surface $z=2x+y^2$ and above the region bounded by $x=y^2$ and $x=y^3$

(c) Under the surface $z=xy$ and above the triangle with vertices $(1,1)$, $(4,1)$ and $(1,2)$

(d) Enclosed by the paraboloid $z=x^2+3y^2$ and the planes $x=0$, $y=1$, $y=x$, $z=0$

(e) Bounded by the cylinder $y^2+z^2=4$ and the planes $x=2y$, $x=0$, $z=0$ in the first octant.

(a)



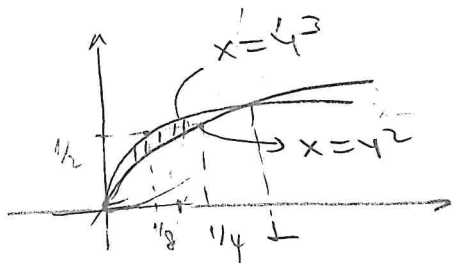
$$y=x \wedge y=x^4$$

$$x=x^4 \quad x(1-x^3)=0 \quad x=0 \quad x=1$$

$$\int_0^1 \int_{x^4}^x (x+2y) dy dx = \int_0^1 (x^4 + y^2) \Big|_{x^4}^x dx = \int_0^1 (x^2 + x^2) - (x^5 + x^8) dx$$

$$= \left[\frac{2}{3} x^3 - \frac{1}{6} x^6 - \frac{1}{9} x^9 \right]_0^1 = \frac{7}{8}$$

(b)

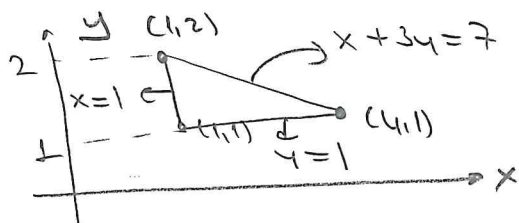


$$\int_0^1 \int_{y^3}^{y^2} (2x+y^2) dx dy =$$

$$\int_0^1 (x^2 + y^2 x) \Big|_{y^3}^{y^2} dy = \int_0^1 (2y^4 - y^6 - y^5) dy$$

$$= \left[\frac{2}{5} y^5 - \frac{1}{7} y^7 - \frac{1}{6} y^6 \right]_0^1 = \frac{19}{210}$$

(c)

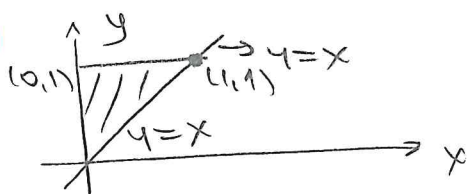


$$V = \int_1^2 \int_1^{7-3y} xy dx dy = \int_1^2 \frac{1}{2} x^2 y \Big|_{x=1}^{x=7-3y} dy$$

$$= \frac{1}{2} \int_1^2 (48y - 42y^2 + 9y^3) dy$$

$$= \frac{1}{2} \left[24y^2 - 14y^3 + \frac{9}{4} y^4 \right]_1^2 = \frac{31}{8}$$

(d)



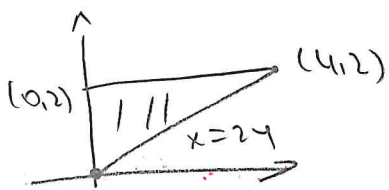
$$V = \int_0^1 \int_x^1 (x^2 + 3y^2) dy dx$$

$$= \int_0^1 x^2 y + y^3 \Big|_x^1 dx$$

$$= \int_0^1 (x^2 + 1) - 2x^3 dx$$

$$= \frac{x^3}{3} + x - 2 \frac{x^4}{4} \Big|_0^1 = \frac{5}{6}$$

(e)



$$V = \int_0^2 \int_0^{2y} \sqrt{4-y^2} dx dy$$

$$= \int_0^2 \sqrt{4-y^2} \cdot 2y dy = -\frac{2}{3} (4-y^2)^{3/2} \Big|_0^2 = \frac{16}{3}$$

(8)

Değer Altınken

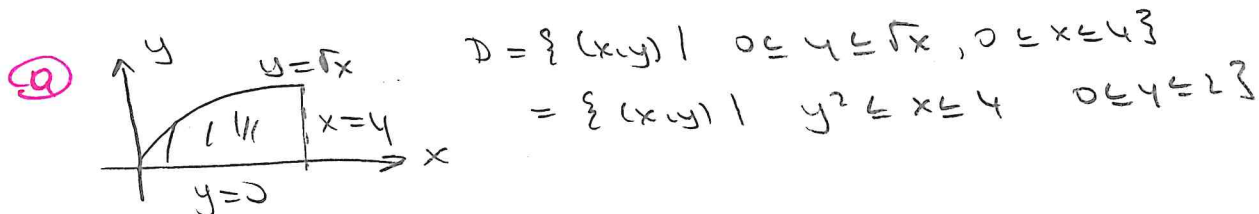
8 Sketch the region of the integrator and change the order

a $\int_0^4 \int_0^{\sqrt{x}} f(x,y) dy dx$

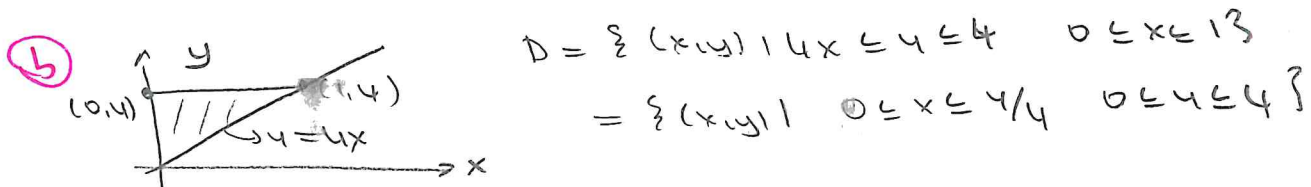
b $\int_0^1 \int_{4x}^4 f(x,y) dy dx$

c $\int_1^2 \int_0^{\ln x} f(x,y) dy dx$

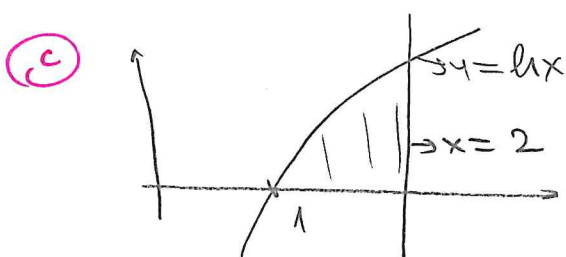
d $\int_0^3 \int_0^{\sqrt{9-y}} f(x,y) dx dy$



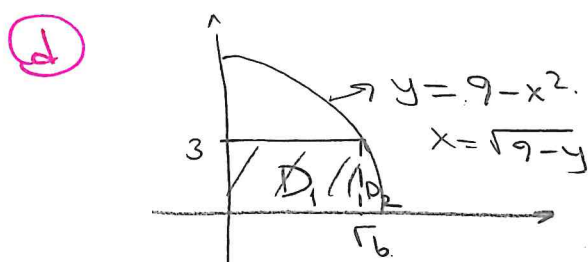
$$\int_0^4 \int_0^{\sqrt{x}} f(x,y) dy dx = \int_0^2 \int_{y^2}^4 f(x,y) dx dy$$



$$\int_0^1 \int_{4x}^4 f(x,y) dy dx = \int_0^4 \int_0^{y/4} f(x,y) dx dy$$



$$\int_1^2 \int_0^{\ln x} f(x,y) dy dx = \int_0^{\ln 2} \int_{e^y}^2 f(x,y) dx dy$$



$$\int_0^3 \int_0^{\sqrt{9-y}} f(x,y) dx dy = \int_0^{\sqrt{6}} \int_0^3 f(x,y) dy dx + \int_{\sqrt{6}}^3 \int_0^{9-x^2} f(x,y) dy dx$$

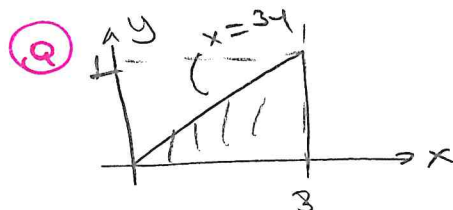
9 Evaluate following integrals by reversing the order of integration.

(a) $\int_0^1 \int_{3y}^3 e^{x^2} dx dy$

(b) $\int_0^1 \int_{\sqrt{y}}^1 \frac{1}{x^3+1} dx dy$

(c) $\int_0^3 \int_{y^2}^9 y \cos(x^2) dx dy$

(d) $\int_0^1 \int_{x^2}^1 x^3 \sin(y^3) dy dx$



$$\int_0^1 \int_{3y}^3 e^{x^2} dx dy = \int_0^3 \int_0^{x/3} e^{x^2} dy dx$$

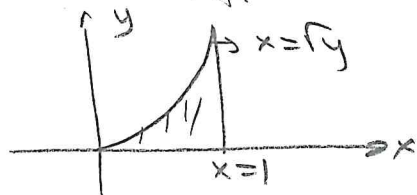
Can't be evaluate in this form

$$= \int_0^3 e^{x^2} y \Big|_0^{x/3} dx = \int_0^3 \frac{x}{3} e^{x^2} dx = \frac{1}{6} e^{x^2} \Big|_0^3 = \frac{e^9 - 1}{6}$$

(b)

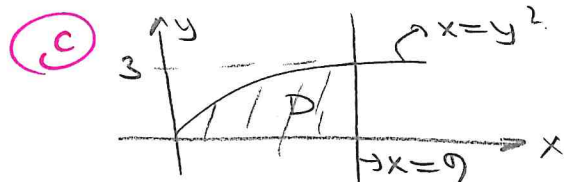
$$\int_0^1 \int_{\sqrt{y}}^1 \frac{1}{x^3+1} dx dy = \int_0^1 \int_0^{x^2} \frac{1}{x^3+1} dy dx$$

Difficult to compute



$$= \int_0^1 \frac{1}{x^3+1} y \Big|_0^{x^2} dx$$

$$= \int_0^1 \frac{1}{x^3+1} \frac{x^2}{\frac{dy}{dx}} = \frac{(x^3+1)^{3/2}}{3/2} \cdot \frac{dy}{3} \Big|_0^1 = \frac{2}{9} (2^{3/2} - 1)$$



$$\int_0^3 \int_{y^2}^9 y \cos(x^2) dx dy = \int_0^9 \int_0^{\sqrt{x}} y \cos(x^2) dy dx$$

$$= \int_0^9 \frac{y^2}{2} \cos(x^2) \Big|_0^{\sqrt{x}} dx$$

$$= \int_0^9 \frac{1}{2} x \cos(x^2) dx \cdot \frac{du}{2} = \frac{1}{4} \sin x^2 \Big|_0^9 = \frac{1}{4} \sin 81$$

10 Estimate the value of integrals (a) $\iint_D \sqrt{x^2+y^2} dA$ $D = [0, 1.3] \times [0, 1.3]$

(b) $I = \iint_D e^{x^2+y^2} dA$ D is the disk with center origin radius $1/2$

(a) $0 \leq \sqrt{x^2+y^2} \leq \sqrt{2}$ so $0 \leq \iint_D \sqrt{x^2+y^2} dA \leq \sqrt{2} \iint_D 1 dA = \sqrt{2} \cdot e^{1/4} \cdot \pi/4$

(b) $D = \{(x,y) \mid x^2+y^2 \leq 1/4\}$ $e^0 \leq e^{x^2+y^2} \leq e^{1/4}$ $\iint_D 1 dA \leq I \leq \iint_D e^{1/4} dA$

(11) Evaluate the given integral by changing to polar coordinates.

(a) $\iint_D xy \, dA$ where D is the disk with center the origin and radius 3

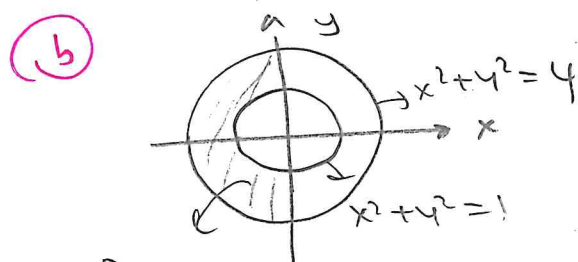
(b) $\iint_R (x+y) \, dA$ where R is the region that lies to the left of the y -axis between the circles $x^2+y^2=1$ and $x^2+y^2=4$

(c) $\iint_R \cos(x^2+y^2) \, dA$ where R is the region that lies above the x axis within the circle $x^2+y^2=9$

(d) $\iint_D e^{x^2-y^2} \, dA$ where D is the region bounded by the semicircle $x=\sqrt{4-y^2}$ and the y -axis

(a) $D = \{ (r, \theta) \mid 0 \leq r \leq 3, 0 \leq \theta \leq 2\pi \}$ $x = r \cos \theta$ $y = r \sin \theta$

$$\iint_D xy \, dA = \int_0^{2\pi} \int_0^3 r \cos \theta \cdot r \sin \theta \cdot r \, dr \, d\theta = \int_0^{2\pi} \cos \theta \sin \theta \left[\frac{r^4}{4} \right]_0^3 d\theta = \frac{81}{4} \int_0^{2\pi} \cos \theta \sin \theta \, d\theta = \frac{81}{4} \left[\frac{\sin^2 \theta}{2} \right]_0^{2\pi} = 0$$



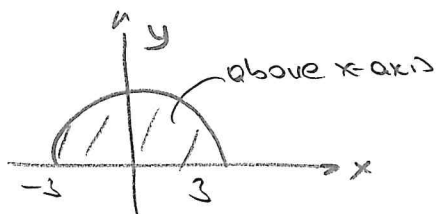
R
left of y axis.

$$\iint_R (x+y) \, dA = \int_{\pi/2}^{3\pi/2} \int_1^2 (r \cos \theta + r \sin \theta) r \, dr \, d\theta = \int_{\pi/2}^{3\pi/2} \left[\frac{r^3}{3} \right]_1^2 (\cos \theta + \sin \theta) d\theta$$

$$= \frac{7}{3} (\sin \theta - \cos \theta) \Big|_{\pi/2}^{3\pi/2} = -\frac{14}{3}$$

Deyan Atkinson

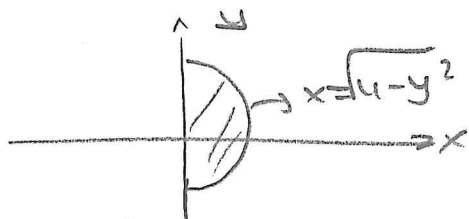
(c)



$$\int_0^\pi \int_0^3 \cos(r^2) r dr d\theta = \int_0^\pi \left. \frac{\sin(r^2)}{2} \right|_0^3 d\theta$$

$$= \int_0^\pi \frac{\sin(9)}{2} d\theta = \left[\frac{\sin \theta}{2} \cdot \pi \right]$$

(d)



$$\iint_D e^{-x^2-y^2} dA = \int_{-\pi/2}^{\pi/2} \int_0^2 e^{-r^2} r dr d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \left. -\frac{e^{-r^2}}{2} \right|_0^2 d\theta$$

$$= (\pi/2 + \pi/2) \cdot \left(-\frac{e^{-4}}{2} + \frac{1}{2} \right)$$

$$= \left(\frac{\pi}{2} \right) (1 - e^{-4})$$

(12) Use polar coordinates to find the volume of the given solid

(9) Under the paraboloid $z = x^2 + y^2$ and above the $x^2 + y^2 \leq 9$

(5) Inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$

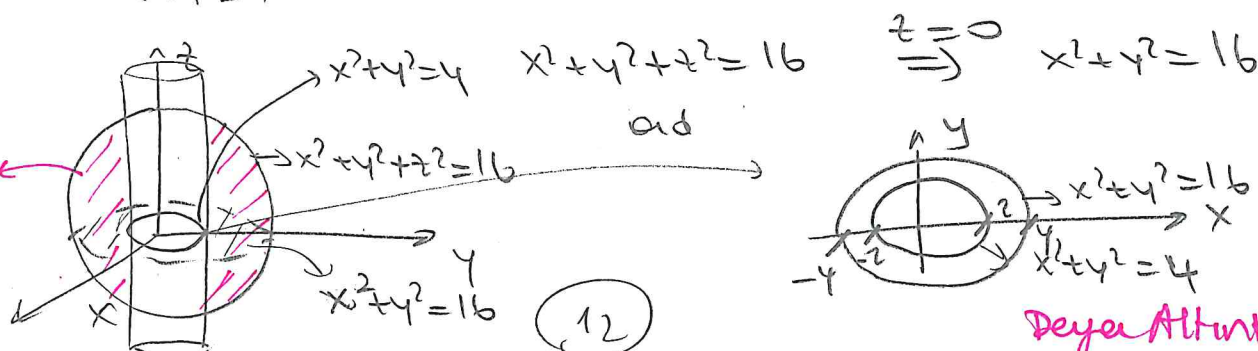
(c) Bounded by the paraboloid $z = 10 - 3x^2 - 3y^2$ and the plane $z = 4$

(d) Above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$

(9) $V = \iint_{x^2+y^2 \leq 9} (x^2+y^2) dA = \int_0^{2\pi} \int_0^3 r^2 \cdot r dr d\theta = \int_0^{2\pi} \left. \frac{r^4}{4} \right|_0^3 d\theta = \left[\frac{81\pi}{2} \right]$

(5)

Solid.



(12)

Deyan Altunhan

$$V = 2 \int_0^{2\pi} \int_2^4 \sqrt{16-x^2-y^2} \, dA = 2 \int_0^{2\pi} \int_2^4 \underbrace{\sqrt{16-r^2}}_u \underbrace{r \, dr \, d\theta}_{-\frac{du}{2}} \, d\theta$$

We have a symmetry

Our solid is $-\sqrt{16-x^2-y^2} \leq z \leq \sqrt{16-x^2-y^2}$

So we use symmetry

$$0 \leq z \leq \sqrt{16-x^2-y^2}$$

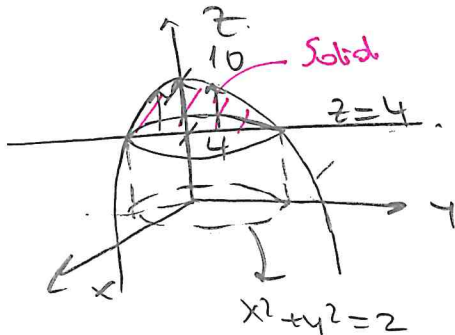
$$-\sqrt{16-x^2-y^2} \leq z \leq 0$$

$$= -\frac{2}{2} \int_0^{2\pi} \frac{(16-r^2)^{3/2}}{3/2} \bigg|_2^4 \, d\theta$$

$$= -\frac{2}{3} \cdot 2\pi (0^{3/2} - (12)^{3/2})$$

$$= \boxed{32\sqrt{3}\pi}$$

(c)



$$z = 10 - 3x^2 - 3y^2 \quad \wedge \quad z = 4$$

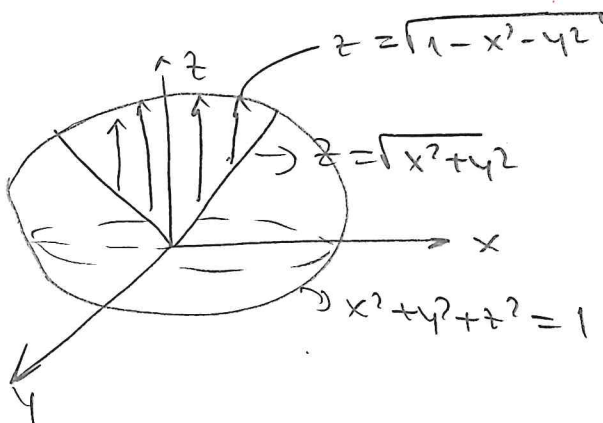
$$10 - 3x^2 - 3y^2 = 4$$

$$x^2 + y^2 = 2$$

$$V = \int_0^{2\pi} \int_0^{\sqrt{2}} ((10 - 3x^2 - 3y^2) - 4) r \, dr \, d\theta$$

$$V = \int_0^{2\pi} \int_0^{\sqrt{2}} (6 - 3r^2) r \, dr \, d\theta = \int_0^{2\pi} (3r^2 - \frac{3}{4}r^4) \bigg|_0^{\sqrt{2}} \, d\theta = \boxed{6\pi}$$

(d)



$$z = \sqrt{x^2 + y^2} \quad \wedge \quad x^2 + y^2 + z^2 = 1$$

$$2(x^2 + y^2) = 1 \quad z = \sqrt{1 - x^2 - y^2}$$

$$x^2 + y^2 = \frac{1}{2}$$

$$\iint (\sqrt{1-x^2-y^2} - \sqrt{x^2+y^2}) \, dA$$

$$x^2 + y^2 \leq \frac{1}{2}$$

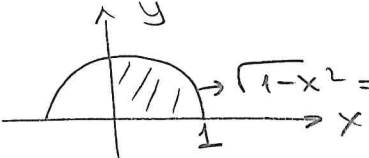
$$\int_0^{2\pi} \int_0^{1/\sqrt{2}} (\sqrt{1-r^2} - r) r \, dr \, d\theta = \int_0^{2\pi} \int_0^{1/\sqrt{2}} \underbrace{\sqrt{1-r^2}}_u \underbrace{r \, dr}_{-\frac{du}{2}} \, d\theta - \int_0^{2\pi} \int_0^{1/\sqrt{2}} r^2 \, dr \, d\theta$$

$$= \int_0^{2\pi} \left(-\frac{1}{3} (1-r^2)^{3/2} - \frac{r^3}{3} \right) \bigg|_0^{1/\sqrt{2}} \, d\theta = 2\pi \left(-\frac{1}{3} \right) \left(\frac{1}{\sqrt{2}} - 1 \right)$$

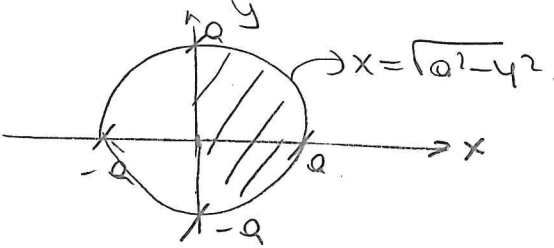
13 Evaluate following integrals by convert to polar coordinates

(a)
$$I = \int_0^1 \int_0^{\sqrt{1-x^2}} e^{x^2+y^2} dy dx$$

(b)
$$I = \int_{-9}^9 \int_0^{\sqrt{9^2-y^2}} (x^2+y^2)^{3/2} dx dy$$

(a) 
$$I = \int_0^{\pi/2} \int_0^1 e^{r^2} r dr d\theta = \frac{1}{2} \int_0^{\pi/2} e^{r^2} \Big|_0^1 d\theta$$

$$= \frac{1}{2} \cdot \frac{\pi}{2} (e-1) = \frac{(e-1)\pi}{4}$$

(b) 
$$I = \int_{-\pi/2}^{\pi/2} \int_0^1 (r^2)^{3/2} r dr d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \frac{r^5}{5} \Big|_0^1 d\theta = \frac{0.5}{5} \cdot (\frac{\pi}{2} + \frac{\pi}{2})$$

$$= \frac{\pi}{5}$$

14 Evaluate the following integrals.

(a)
$$\int_0^1 \int_0^7 \int_0^{x+z} 6xz dy dx dz$$

(b)
$$\int_0^3 \int_0^1 \int_0^{\sqrt{1-z^2}} ze^y dx dz dy$$

(c)
$$\iiint_E 2x dv$$

$$E = \{(x,y,z) \mid 0 \leq y \leq 2, 0 \leq x \leq \sqrt{4-y^2}, 0 \leq z \leq 4\}$$

(d)
$$\iiint_E yz \cos(xz) dv$$

$$E = \{(x,y,z) \mid 0 \leq x \leq 1, 0 \leq y \leq x, x \leq z \leq 2x\}$$

(e)
$$\iiint_E y dv$$
 where E is bounded by the planes $x=0$, $y=0$, $z=0$, $2x+2y+z=4$

(f)
$$\iiint_E x^2 e^y dv$$
 where E is bounded by the parabolic cylinder $z=1-y^2$ and the planes $z=0$, $x=1$ and $x=-1$

Deyan Atkinson

$$(a) \int_0^1 \int_0^z \int_0^{x+z} 6x + 4y \, dx \, dz = \int_0^1 \int_0^z [6xy + 2y^2] \Big|_0^{x+z} dx \, dz =$$

$$= \int_0^1 \int_0^z 6x + 4z(x+z) \, dx \, dz = \int_0^1 [2x^2 + 3x^2z^2] \Big|_{x=0}^{x=z} dz = \int_0^1 (2z^4 + 3z^4) dz$$

$$= \int_0^1 5z^4 dz = z^5 \Big|_0^1 = 1$$

$$(b) \int_0^3 \int_0^1 \int_0^{\sqrt{1-z^2}} z \cdot e^y \, dx \, dz \, dy = \int_0^3 \int_0^1 z \cdot e^y \cdot x \Big|_0^{\sqrt{1-z^2}} dz \, dy =$$

$$= \int_0^3 \int_0^1 z \cdot e^y (\sqrt{1-z^2} - 0) \, dz \, dy = \int_0^3 \int_0^1 e^y \cdot \sqrt{1-z^2} \cdot z \, dz \, dy$$

$$= \int_0^3 \left[-\frac{e^y}{2} \frac{(1-z^2)^{3/2}}{3/2} \right]_0^1 dy = \int_0^3 \frac{1}{3} e^y dy = \frac{1}{3} e^y \Big|_0^3 = \frac{1}{3} (e^3 - 1)$$

$$(c) \iiint_E 2x \, dx = \int_0^2 \int_0^{\sqrt{4-y^2}} \int_0^y 2x \, dz \, dx \, dy = \int_0^2 \int_0^{\sqrt{4-y^2}} 2x \cdot z \Big|_{z=0}^y dx \, dy$$

$$= \int_0^2 \int_0^{\sqrt{4-y^2}} 2xy \, dx \, dy = \int_0^2 x^2 y \Big|_0^{\sqrt{4-y^2}} dy = \int_0^2 (4-y^2) \cdot y \, dy$$

$$= \int_0^2 4y - y^3 = 2y^2 - \frac{y^4}{4} \Big|_0^2 = 4$$

$$(d) \iiint_E yz \cos(x^5) \, dV = \int_0^1 \int_0^x \int_0^{2x} yz \cos(x^5) \, dz \, dy \, dx$$

$$= \int_0^1 \int_0^x y \cdot \frac{z^2}{2} \cos(x^5) \Big|_0^{2x} dy \, dx = \int_0^1 \int_0^x \frac{3x^2}{2} y \cdot \cos(x^5) dy \, dx$$

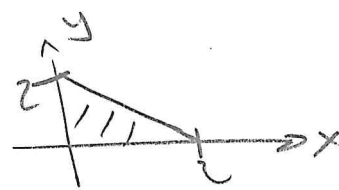
$$= \int_0^1 \frac{3x^2}{2} \cos(x^5) \cdot \frac{y^2}{2} \Big|_0^x dx = \frac{3}{4} \int_0^1 x^4 \cos(x^5) dx$$

$$= \frac{3}{4} \frac{\sin(x^5)}{5} \Big|_0^1 = \frac{3}{20} \sin 1$$

Değer Altınken

e)
$$\iiint_E y \, dV = \int_0^2 \int_0^{2-x} \int_0^{4-2x-2y} y \, dz \, dy \, dx = \int_0^2 \int_0^{2-x} y(4-2x-2y) \, dy \, dx$$

$2x+2y+z=4 \xrightarrow{z=0} 2x+2y=4 \quad (x+y=2)$
Domain D



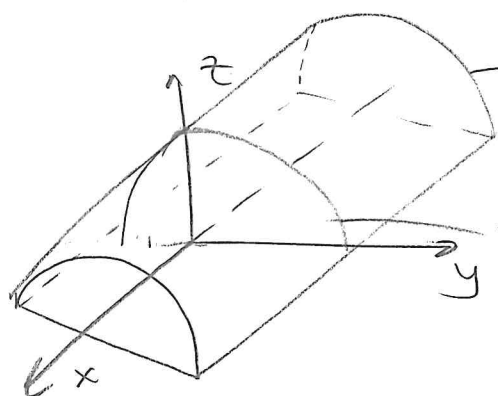
$$= \int_0^2 \int_0^{2-x} (4y - 2xy - 2y^2) \, dy \, dx = \int_0^2 \left[2y^2 - xy^2 - \frac{2}{3}y^3 \right] \Big|_{y=0}^{y=2-x} \, dx$$

$$= \int_0^2 2(2-x)^2 - x(2-x)^2 - \frac{2}{3}(2-x)^3 \, dx$$

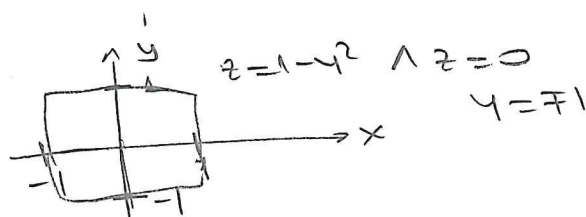
$$= \int_0^2 (2-x)^2 [2-x] - \frac{2}{3}(2-x)^3 \, dx$$

$$= \int_0^2 \frac{1}{3}(2-x)^3 \, dx = -\frac{1}{3} \frac{(2-x)^4}{4} \Big|_0^2 = \frac{4}{3}$$

f)



$z = 1 - y^2$
 $\int_{-1}^1 \int_{-1}^1 \int_0^{1-y^2} x^2 e^y \, dz \, dy \, dx$



$$\int_{-1}^1 \int_{-1}^1 x^2 e^y (1-y^2) \, dy \, dx = \int_{-1}^1 \int_{-1}^1 x^2 (e^y - y^2 e^y) \, dy \, dx$$

$$= \int_{-1}^1 x^2 (e^y - (y^2 - 2y + 2)e^y) \Big|_{-1}^1 \, dx$$

$$\int y^2 e^y \, dy = y^2 e^y - 2 \int y e^y \, dy = y^2 e^y - 2(y e^y - e^y)$$

$u = y^2 \Rightarrow du = 2y \, dy$

$y = u \Rightarrow e^y \, dy = du$

$\Downarrow$
 $dy = du \quad e^y = u$

$$= \frac{1}{3} [e - e - e^{-1} + 5e^{-1}] = \frac{4}{3}e$$

Değer Altınlar

1b)

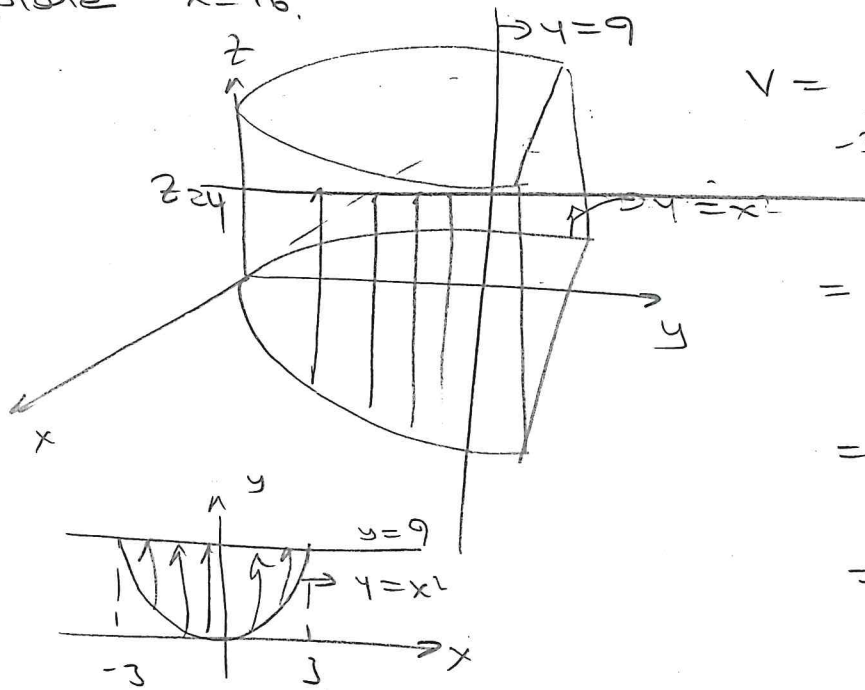
15 Use triple integral to find the volume of the given solid

a The solid bounded by ^{parabolic cylinder} $y = x^2$ and the planes $z = 0, z = 4$ and $y = 9$

b The solid enclosed by the cylinder $x^2 + y^2 = 9$ and the planes $y + z = 5$ and $z = 1$

c The solid enclosed by the paraboloid $x = y^2 + z^2$ and the plane $x = 16$.

c



$$V = \int_{-3}^3 \int_{x^2}^9 \int_0^4 dz dy dx$$

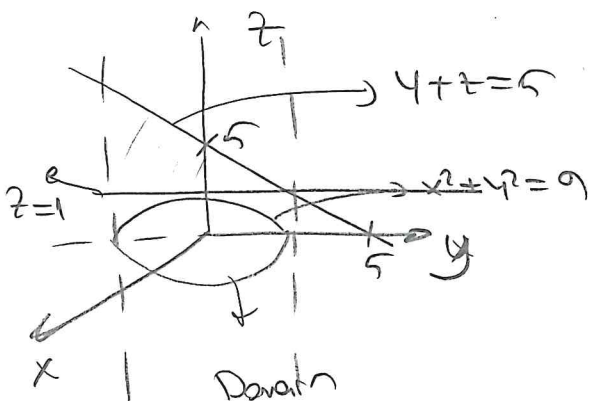
$$= 4 \int_{-3}^3 \int_{x^2}^9 dy dx$$

$$= 4 \int_{-3}^3 (9 - x^2) dx$$

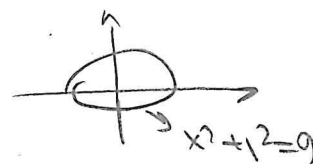
$$= 4 \cdot \left(9x - \frac{x^3}{3} \right) \Big|_{-3}^3$$

$$= 4(27 - 9 + 27 - 9) = 144$$

b



$$I = \int_{-3}^3 \int_{\sqrt{9-x^2}}^{5-y} \int_1^5 dz dy dx$$



I. copy

$$I = \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (5 - y - 1) dy dx = \int_{-3}^3 \left(4y - \frac{y^2}{2} \right) \Big|_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx$$

$$= \int_{-3}^3 8\sqrt{9-x^2} dx \Rightarrow \text{Need trigonometric substitution}$$

(17)

Deja Attention

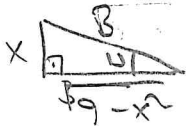
$$\int \sqrt{9-x^2} dx = \int 3 \cos u \sqrt{9-9\sin^2 u} du = \int 9 \cos^2 u du$$

$$x = 3 \sin u \Rightarrow u = \arcsin\left(\frac{x}{3}\right) = \int 9 \frac{\cos 2u + 1}{2}$$

$$dx = 3 \cos u$$

$$\cos^2 u = \frac{\cos 2u + 1}{2}$$

$$\sin u = \frac{x}{3}$$



$$\begin{aligned} \sin 2u &= 2 \sin u \cos u \\ &= 2 \frac{x}{3} \frac{\sqrt{9-x^2}}{3} \end{aligned}$$

$$I = 36\pi$$

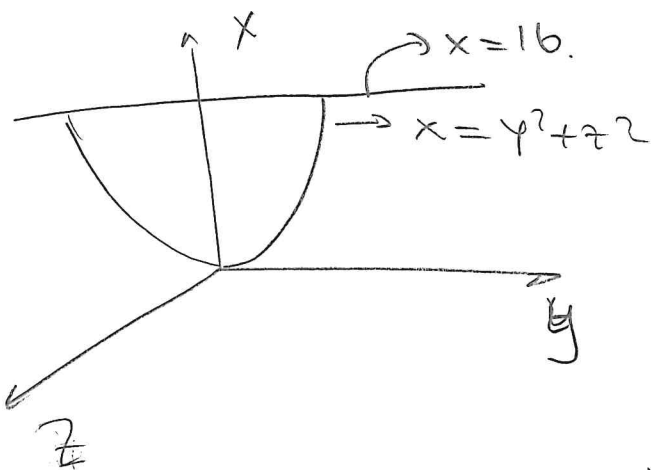
II. way

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (4-y) dy dx = \int_0^{2\pi} \int_0^3 (4-r \sin \theta) r dr d\theta$$

$$= \int_0^{2\pi} \left(2r^2 - \frac{r^3}{3} \sin \theta \right) \Big|_{r=0}^3 d\theta = \int_0^{2\pi} (18 - 9 \sin \theta) d\theta$$

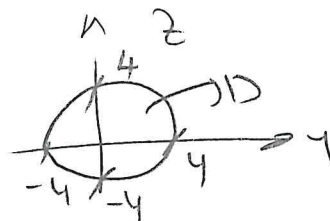
$$= 18\theta + 9 \cos \theta \Big|_0^{2\pi} = 36\pi$$

(c)



$$x=16 \wedge x=y^2+z^2$$

$$y^2+z^2=16$$



$$\iint_D \int_{x=y^2+z^2}^{x=16} dx dA = \int_0^{2\pi} \int_0^4 (16 - \underbrace{y^2+z^2}_{r^2}) r dr d\theta$$

$$= 2\pi \left(16 \frac{r^2}{2} - \frac{r^4}{4} \right) \Big|_0^4 = 128\pi = \int_0^{2\pi} \int_0^4 (16-r^2) r dr d\theta$$

Duyet Alifintan

15 Express the integral $\iiint_E f(x,y,z) dV$ where E is the solid bounded by the given surfaces.

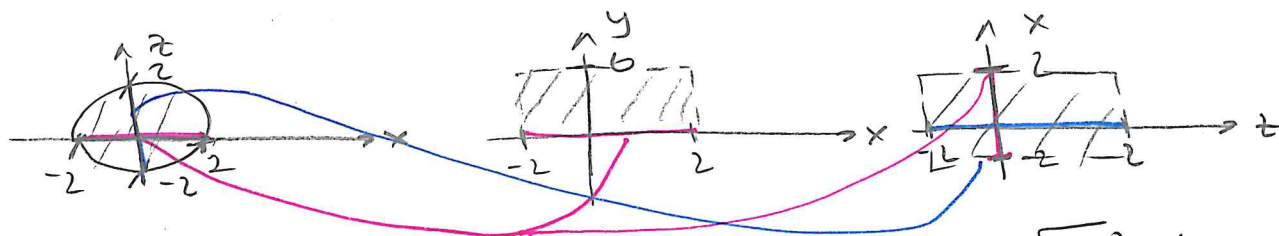
(a) $x^2 + z^2 = 4$ $y=0$ $y=6$

(b) $z=0$ $x=0$ $y=2$ $z=y-2x$

(c) $z=0$ $z=y$ $x^2=1-y$

(d) $9x^2 + 4y^2 + z^2 = 1$

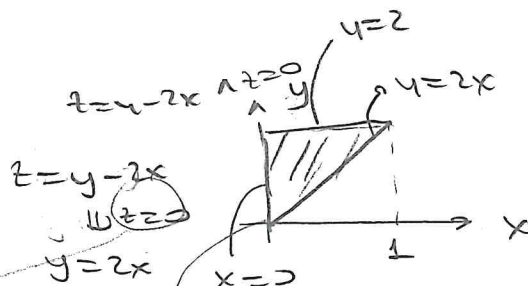
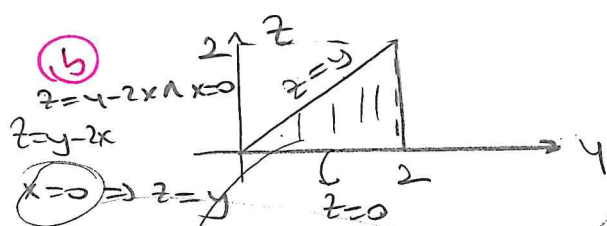
(a)



$$\int_0^6 \int_{-2}^2 \int_{-\sqrt{4-z^2}}^{\sqrt{4-z^2}} f(x,y,z) dx dz dy = \int_{-2}^2 \int_0^6 \int_{-\sqrt{4-z^2}}^{\sqrt{4-z^2}} f(x,y,z) dx dy dz = \int_{x=-2}^2 \int_{z=\sqrt{4-x^2}}^6 \int_{y=0}^6 f(x,y,z) dy dz dx$$

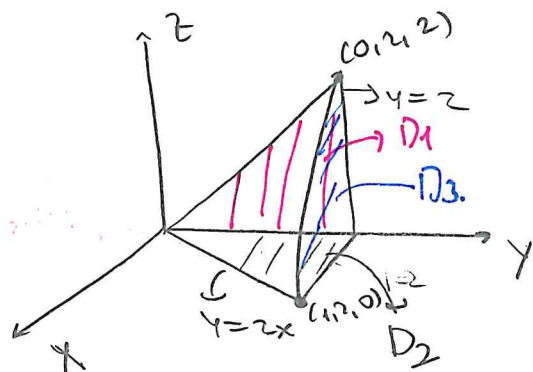
$$= \int_{-2}^2 \int_{-\sqrt{4-z^2}}^{\sqrt{4-z^2}} \int_0^6 f(x,y,z) dy dz dx$$

(b)

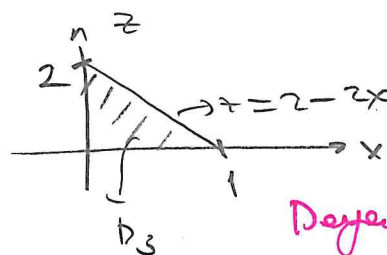


$$D_2 = \{(x,y) \mid 0 \leq x \leq 1, 2x \leq y \leq 2\}$$

$$D_1 = \{(y,z) \mid 0 \leq y \leq 2, 0 \leq z \leq y\}$$



Since $2x \leq y \leq 2$
 $2x - 2x \leq y - 2x \leq 2 - 2x$
 $z = y - 2x \leq 2 - 2x$



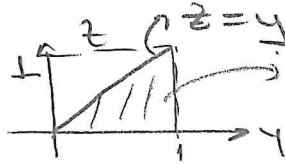
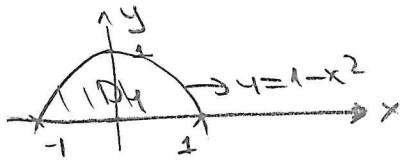
Pay Attention

$$= \int_0^1 \int_0^{2-2x} \int_{2x}^{2-2x} f(x,y,z) dy dz dx$$

$$\iiint_E f(x,y,z) dV = \int_0^1 \int_{2x}^2 \int_0^{y-2x} f(x,y,z) dz dy dx = \int_0^1 \int_0^{y/2} \int_0^{y-2x} f(x,y,z) dz dy dx$$

$$= \int_0^2 \int_0^y \int_0^{(y-z)/2} f(x,y,z) dx dz dy = \int_0^2 \int_0^2 \int_0^{(y-z)/2} f(x,y,z) dx dy dz$$

c) $z=0$ $z=y$ $x^2=1-y$



$D_2 = \{(x,y) | 0 \leq y \leq 1, 0 \leq z \leq y\}$

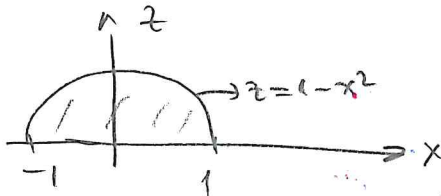
$D_1 = \{(x,y) | -1 \leq x \leq 1, 0 \leq y \leq 1-x^2\}$

$-y \leq -z$

$1-y \leq 1-z$

$x^2 \leq 1-y \leq 1-z \Rightarrow z \leq 1-x^2$

$D_3 = \{(x,z) | -1 \leq x \leq 1, 0 \leq z \leq 1-x^2\}$



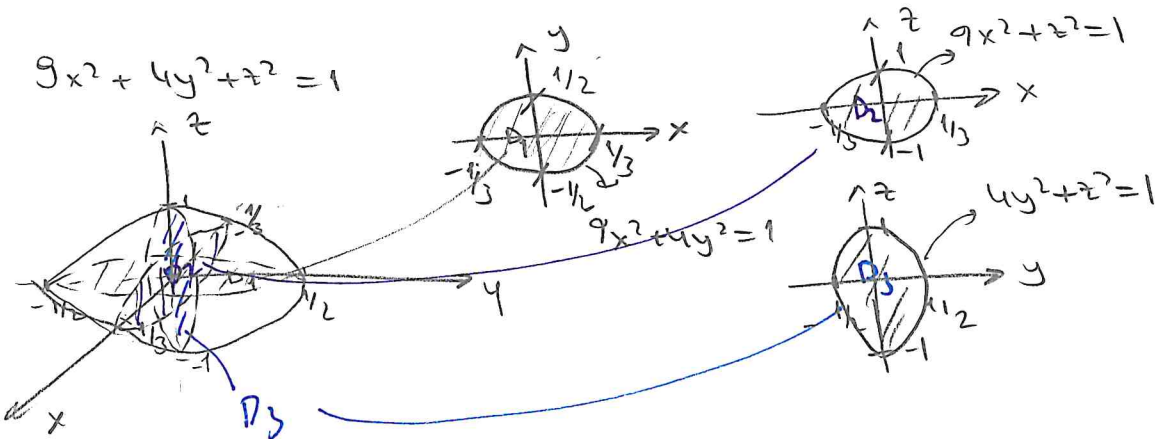
$$\iiint_E f(x,y,z) dV = \int_{-1}^1 \int_0^{1-x^2} \int_0^y f(x,y,z) dz dy dx = \int_0^1 \int_{-\sqrt{1-y}}^{\sqrt{1-y}} \int_0^y f(x,y,z) dz dx dy$$

$$= \int_0^1 \int_{-\sqrt{1-y}}^{\sqrt{1-y}} \int_0^y f(x,y,z) dx dz dy = \int_0^1 \int_{-\sqrt{1-y}}^{\sqrt{1-y}} \int_z^{1-x^2} f(x,y,z) dy dz dx = \int_{-1}^1 \int_0^{1-x^2} \int_z^{1-x^2} f(x,y,z) dy dz dx$$

$$= \int_0^1 \int_{-\sqrt{1-z}}^{\sqrt{1-z}} \int_z^{1-x^2} f(x,y,z) dy dx dz$$

d)

$9x^2 + 4y^2 + z^2 = 1$



$$\iiint_E f(x,y,z) dV = \int_{-1/3}^{1/3} \int_{-\sqrt{1-9x^2}}^{\sqrt{1-9x^2}} \int_0^{\sqrt{1-9x^2-4y^2}} f(x,y,z) dz dy dx$$

$$= \int_{-1/2}^{1/2} \int_{-\sqrt{1-4y^2}}^{\sqrt{1-4y^2}} \int_0^{\sqrt{1-9x^2-4y^2}} f(x,y,z) dz dx dy$$

$$= \int_{-1/2}^{1/2} \int_{-\sqrt{1-4y^2}}^{\sqrt{1-4y^2}} \int_{\sqrt{1-4y^2-z^2}/3}^{\sqrt{1-4y^2-z^2}/3} f(x,y,z) dx dz dy$$

$$= \int_{-1}^1 \int_{-\sqrt{1-z^2}/2}^{\sqrt{1-z^2}/2} \int_{-\sqrt{1-4y^2-z^2}/3}^{\sqrt{1-4y^2-z^2}/3} f(x,y,z) dy dz dx$$

Deja Attin

16) Evaluate the following integrals.

a) $\iiint_E \sqrt{x^2 + y^2} \, dV$ where E is the region that lies the cylinder $x^2 + y^2 = 16$ between the planes $z = -5$ and $z = 4$

b) $\iiint_B (x^2 + y^2 + z^2) \, dV$ where B is the unit ball $x^2 + y^2 + z^2 \leq 1$

c) $\iiint_E (x^3 + xy^2) \, dV$ where E lies in the first octant that lies inside the paraboloid $z = 1 - x^2 - y^2$

d) $\iiint_H (x^2 + y^2) \, dV$ where H is the hemispherical region that lies above the xy -plane and below the sphere $x^2 + y^2 + z^2 = 1$

e) Find volume of the solid that lies within both the cylinder $x^2 + y^2 = 4$ and the sphere $x^2 + y^2 + z^2 = 4$

f) Find the volume of the solid that lies above the cone $\phi = \pi/3$ and below the sphere $\rho = 4 \cos \phi$

g) $E = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 4, -5 \leq z \leq 4\}$

$$\begin{aligned} \iiint_E \sqrt{r^2} \cdot r \, dz \, dr \, d\theta &= \int_0^{2\pi} \int_0^4 \int_{-5}^4 r^2 \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^4 9r^2 \, dr \, d\theta \\ &= \int_0^{2\pi} 3r^3 \Big|_0^4 \, d\theta = 192 \cdot 2\pi = \underline{384\pi} \end{aligned}$$

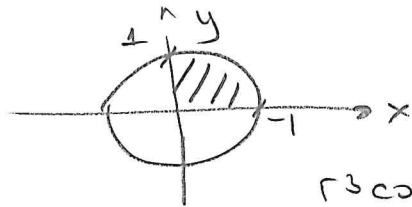
h) $B = \{(\rho, \phi, \theta) \mid 0 \leq \rho \leq 1, 0 \leq \phi \leq \pi, 0 \leq \theta \leq 2\pi\}$

$$\begin{aligned} \iiint_B \rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta &= \int_0^{2\pi} \int_0^\pi \int_0^1 \rho^5 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= \frac{1}{6} \int_0^{2\pi} -\cos \phi \Big|_0^\pi \, d\theta = \underline{\frac{4\pi}{3}} \end{aligned}$$

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c) $z = 1 - x^2 - y^2$ intersect xy plane in the circle $x^2 + y^2 = 1$ or $r = 1$
 first octant $x \geq 0, y \geq 0, z \geq 0$

$$E = \{ (r, \theta, z) \mid 0 \leq \theta \leq \pi/2, 0 \leq r \leq 1, 0 \leq z \leq \frac{1-r^2}{1-(x^2+y^2)} \}$$



$$\iiint_E (x^3 + xy^2) dV = \int_0^{\pi/2} \int_0^1 \int_0^{1-r^2} (r^3 \cos^3 \theta + r^5 \cos \theta \sin^2 \theta) r dz dr d\theta$$

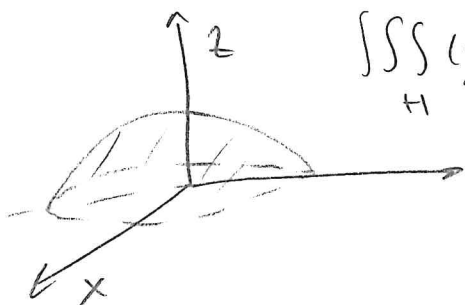
$$= \int_0^{\pi/2} \int_0^1 r^4 \cos \theta (1-r^2) dr d\theta = \int_0^{\pi/2} \cos \theta \left[\frac{1}{5} r^5 - \frac{1}{7} r^7 \right]_{r=0}^{r=1} d\theta$$

$$= \int_0^{\pi/2} \cos \theta \left[\frac{1}{5} - \frac{1}{7} \right] d\theta = \int_0^{\pi/2} \frac{2}{35} \cos \theta d\theta = \frac{2}{35}$$

$$= \int_0^{\pi/2} \frac{2}{35} \cos \theta d\theta = \frac{2}{35} \sin \theta \Big|_0^{\pi/2} = \frac{2}{35}$$

$$= \frac{2}{35}$$

d) $H = \{ (\rho, \phi, \theta) \mid 0 \leq \rho \leq 1, 0 \leq \phi \leq \pi/2, 0 \leq \theta \leq 2\pi \}$



$$\iiint_H (x^2 + y^2) dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 (\rho^2 \sin^2 \phi) \rho^2 \sin \phi d\rho d\phi d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/2} \rho^5 \sin^3 \phi d\phi d\theta$$

$$= \int_0^{2\pi} \left[-\cos \phi + \frac{1}{3} \cos^3 \phi \right]_0^{\pi/2} d\theta$$

$$= \int_0^{2\pi} \left[-1 + \frac{1}{3} \right] d\theta = \int_0^{2\pi} -\frac{2}{3} d\theta = -\frac{4\pi}{3}$$

$$= -\frac{4\pi}{3}$$

$$= \frac{4\pi}{3}$$

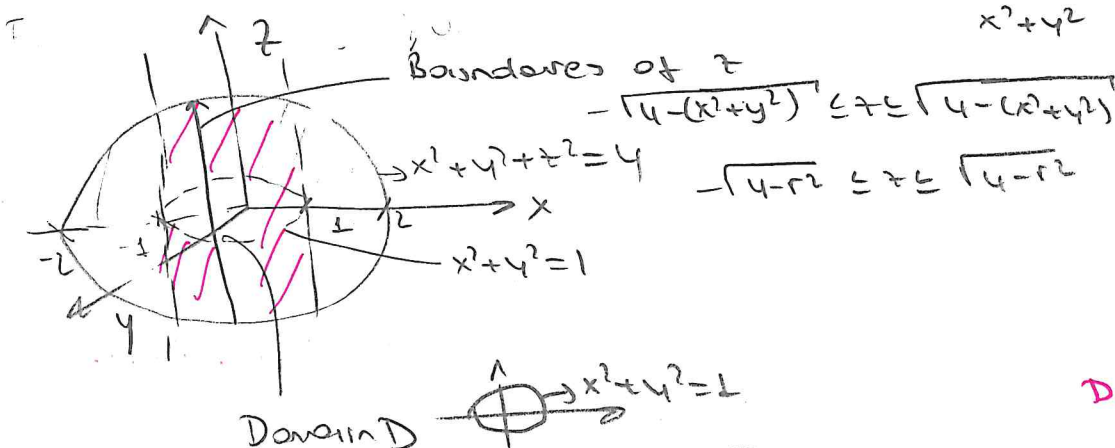
$$\int \sin^3 \phi d\phi = \int (1 - \cos^2 \phi) \sin \phi d\phi$$

$$\cos \phi = u$$

$$-\sin \phi d\phi = -du = -u^2 + \frac{u^3}{3}$$

$$= -\cos \phi + \frac{1}{3} \cos^3 \phi$$

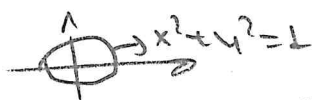
e) $E = \{ (r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 1, -\sqrt{4-r^2} \leq z \leq \sqrt{4-r^2} \}$



Boundaries of z

$$-\sqrt{4-(x^2+y^2)} \leq z \leq \sqrt{4-(x^2+y^2)}$$

$$-\sqrt{4-r^2} \leq z \leq \sqrt{4-r^2}$$



Darya Altıntaş

$$\iiint_E dV = \int_0^{2\pi} \int_0^1 \int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^1 2r\sqrt{4-r^2} \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[-\frac{2}{3} (4-r^2)^{3/2} \right]_0^1 d\theta = \frac{4}{3} \pi (8 - 3^{3/2})$$

(+) Since $\rho = 4 \cos \phi$ implies $\rho^2 = 4\rho \cos \phi$ the equation of the sphere with radius 2 with center (0,0,2)

$x^2 + y^2 + z^2 = 4x$ $\times$ $x^2 + y^2 + z^2 - 4x = 0$
 $(x^2 - 4x + 4) + y^2 + z^2 = 4$
 $(x-2)^2 + y^2 + z^2 = 4$

$$V = \int_0^{2\pi} \int_0^{\pi/3} \int_0^{4 \cos \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/3} \frac{1}{3} \rho^3 \Big|_0^{4 \cos \phi} \sin \phi \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/3} \left(\frac{64}{3} \cos^3 \phi \right) \sin \phi \, d\phi \, d\theta = \int_0^{2\pi} \left[-\frac{16}{3} \cos^4 \phi \right]_0^{\pi/3} d\theta$$

$$= \int_0^{2\pi} -\frac{16}{3} \left(\frac{1}{16} - 1 \right) d\theta = 5\theta \Big|_0^{2\pi} = 10\pi$$

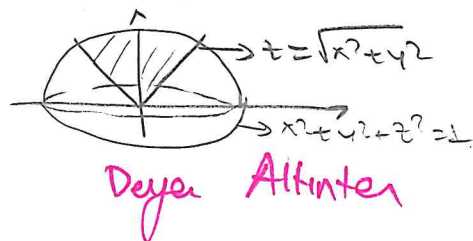
(17) Write the following integrals in (a) spherical (b) cylindrical (c) rectangular (d) Evaluate the integral by using one of (a), (b), (c)

(18) Volume of the solid E that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$

(19) $\iiint_E dV$ where E lies above the paraboloid $z = x^2 + y^2$ and below the plane $z = 2y$ (only (b) and (c))

(1)

(a) Spherical $z = \sqrt{x^2 + y^2}$ $x^2 + y^2 = \rho^2 \sin^2 \phi$ $z^2 = \rho^2 \cos^2 \phi$
 $\rho \cos \phi = \rho \sin \phi \Rightarrow \phi = \pi/4$



$$x^2 + y^2 + z^2 = \rho^2 = 1 \quad \rho = 1$$

$$\int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

5

Cylindrical

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

$$z = \sqrt{x^2 + y^2} = r^2$$

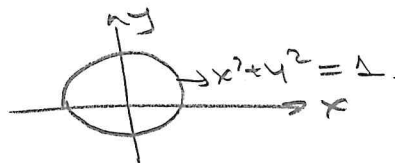
$$x^2 + y^2 + z^2 = 1 \quad z = \sqrt{1 - r^2}$$

$$\int_0^{2\pi} \int_0^{1/2} \int_{r^2}^{\sqrt{1-r^2}} r \, dz \, dr \, d\theta$$

$$z = \sqrt{x^2 + y^2} \quad \wedge \quad z^2 + x^2 + y^2 = 1$$

$$x^2 + y^2 + x^2 + y^2 = 1$$

$$x^2 + y^2 = 1/2$$



6

Rectangular

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{x^2+y^2}}^{\sqrt{x^2+y^2}} dz \, dy \, dx$$

1635

7

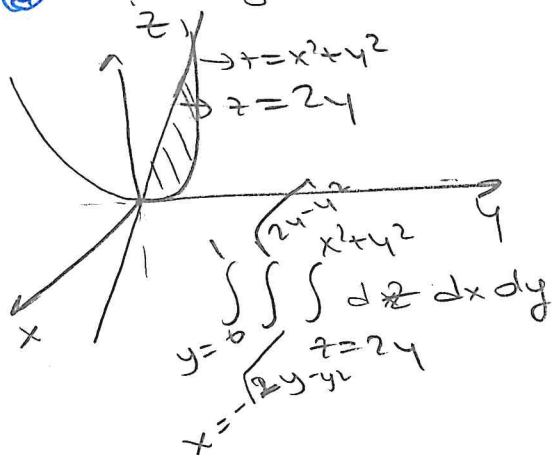
Spherical

$$V = \int_0^{2\pi} \int_0^{\pi/4} \int_0^1 r^2 \sin \phi \, dr \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} \frac{r^3}{3} \Big|_0^1 \sin \phi \, d\phi \, d\theta$$

$$= \frac{1}{3} \int_0^{2\pi} \int_0^{\pi/4} \sin \phi \, d\phi \, d\theta = \frac{1}{3} \int_0^{2\pi} (-\cos \phi) \Big|_0^{\pi/4} d\theta = \frac{1}{3} \pi (2 - \sqrt{2})$$

8

Rectangular



$$z = 2y \quad \wedge \quad z = x^2 + y^2$$

$$x^2 + y^2 = 2y$$

$$x^2 + y^2 - 2y + 1 = 1$$

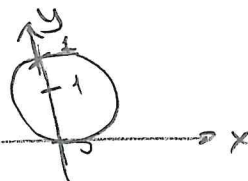
$$x^2 + (y-1)^2 = 1$$

$$x^2 = 1 - (y-1)^2$$

$$x^2 = 1 - y^2 + 2y - 1$$

$$\Rightarrow x^2 = 2y - y^2$$

$$x = \pm \sqrt{2y - y^2}$$



9

Cylindrical

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$V = \int_0^{2\pi} \int_0^{2\pi \sin \theta} \int_{2r \sin \theta}^{\sqrt{1-r^2}} r \, dz \, dr \, d\theta =$$

$$z = 2y \Rightarrow z = 2r \sin \theta \quad \left. \begin{array}{l} z = 2r \sin \theta \\ z = \sqrt{1-r^2} \end{array} \right\} \quad \begin{array}{l} z = 2r \sin \theta \wedge z = \sqrt{1-r^2} \\ 2r \sin \theta = \sqrt{1-r^2} \\ r = 2 \sin \theta \end{array}$$

10

Spherical

Ex 8) Evaluate the following integrals by changing to cylindrical coordinates

a)
$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{x^2+y^2}^{2-x^2-y^2} (x^2+y^2)^{3/2} dz dy dx$$

b)
$$\int_0^1 \int_0^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xy z dz dx dy$$

a)
$$\begin{aligned} x^2+y^2 \leq z \leq 2-x^2-y^2 \\ -\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2} \\ -1 \leq x \leq 1 \end{aligned} \Rightarrow x^2+y^2=1$$

Cylindrical Coordinates
 $x = r \cos \theta$
 $y = r \sin \theta$
 $x^2+y^2 = r^2$

$r^2 \leq z \leq 2-r^2$
 $0 \leq \theta \leq 2\pi$
 $0 \leq r \leq 1$

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{x^2+y^2}^{2-x^2-y^2} (x^2+y^2)^{3/2} dz dy dx =$$

$$\int_0^{2\pi} \int_0^1 \int_{r^2}^{2-r^2} (r^2)^{3/2} r dz dr d\theta =$$

$$= \int_0^{2\pi} \int_0^1 r^4 z \Big|_{r^2}^{2-r^2} dr d\theta = \int_0^{2\pi} \int_0^1 r^4 (2-r^2-r^2) dr d\theta =$$

$$= \int_0^{2\pi} 2 \frac{r^5}{5} \frac{2r^7}{7} \Big|_0^1 d\theta = 2\pi \left(\frac{4}{35} \right) = \frac{8\pi}{35}$$

b)
$$\int_0^1 \int_0^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xy z dz dx dy$$

$$\begin{aligned} x^2+y^2 \leq z \leq \sqrt{x^2+y^2} \\ 0 \leq x \leq \sqrt{1-y^2} \\ 0 \leq y \leq 1 \end{aligned} \Rightarrow \begin{aligned} r^2 \leq z \leq r \\ x^2+y^2=1 \end{aligned}$$

$$\begin{aligned} z &= z \\ y &= r \sin \theta \\ x &= r \cos \theta \\ x^2+y^2 &= r^2 \end{aligned}$$

$$\int_0^{\pi/2} \int_0^1 \int_{r^2}^r r \cos \theta r \sin \theta z \cdot r dz dr d\theta = \int_0^{\pi/2} \int_0^1 r^3 \cos \theta \sin \theta \cdot \frac{z^2}{2} \Big|_{r^2}^r dr d\theta$$

$$= \int_0^{\pi/2} \int_0^1 \frac{r^3}{2} (r^2 - r^4) \cos \theta \sin \theta dr d\theta = \int_0^{\pi/2} \frac{1}{2} \left(\frac{r^6}{6} - \frac{r^8}{8} \right) \Big|_0^1 \cos \theta \sin \theta d\theta$$

$$= \frac{1}{2} \left[-\frac{1}{2} \cos 2\theta \right]_0^{\pi/2} = \frac{1}{2}$$

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19) Evaluate the following integral by changing to spherical coordinates

(a)
$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2-y^2}} \frac{1}{\sqrt{x^2+y^2+z^2}} dz dy dx$$

(b)
$$\int_0^3 \int_0^{\sqrt{9-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{18-x^2-y^2}} (x^2+y^2+z^2) dz dx dy$$

(a)
$$\begin{aligned} x &= \rho \sin \phi \cos \theta \\ y &= \rho \sin \phi \sin \theta \\ z &= \rho \cos \phi \end{aligned}$$

$$x^2+y^2+z^2 = \rho^2 \quad z \cdot \sqrt{x^2+y^2+z^2} = \rho^2 \cos \phi$$

$$dV = \rho^2 \sin \phi d\rho d\phi d\theta$$

$$x^2+y^2 = \rho^2 \sin^2 \phi$$

$$z^2 = \rho^2 \cos^2 \phi$$

$$0 \leq z \leq \sqrt{9-x^2-y^2} \Rightarrow$$

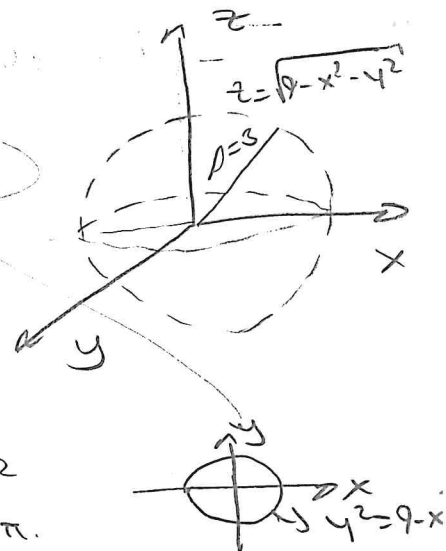
$$-\sqrt{9-x^2} \leq y \leq \sqrt{9-x^2}$$

$$-3 \leq x \leq 3$$

$$0 \leq \rho \leq 3$$

$$0 \leq \phi \leq \pi/2$$

$$0 \leq \theta \leq 2\pi$$



$$\int_0^{2\pi} \int_0^{\pi/2} \int_0^3 \rho^2 \sin \phi \rho^2 \sin \phi d\rho d\phi d\theta = \int_0^{2\pi} \int_0^{\pi/2} \frac{\rho^5}{5} \Big|_0^3 \sin \phi \cos \phi d\phi d\theta$$

$$= \frac{243}{5} \int_0^{2\pi} \frac{\sin^2 \phi}{2} \Big|_0^{\pi/2} d\theta = \frac{1}{2} \cdot \frac{243}{5} \cdot 2\pi = \frac{243}{5} \pi$$

(b)
$$\sqrt{x^2+y^2} \leq z \leq \sqrt{18-x^2-y^2}$$

$$0 \leq x \leq \sqrt{9-y^2}$$

$$\pi/4 \leq \theta \leq \pi/2 \quad \sqrt{2} \leq y \leq 3$$

$$\int_0^{\pi/2} \int_{\pi/4}^{\pi/2} \int_0^{\sqrt{18-x^2-y^2}} \rho^2 \sin \phi d\rho d\phi d\theta = \int_0^{\pi/2} \int_{\pi/4}^{\pi/2} \frac{\rho^5}{5} \sin \phi d\phi d\theta$$

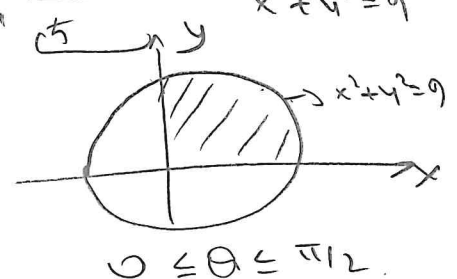
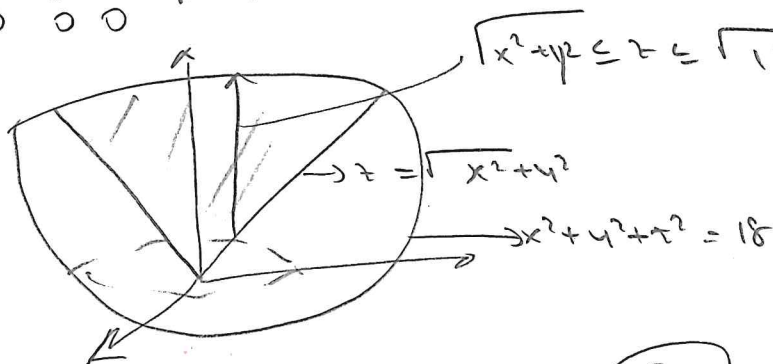
$$z = \sqrt{x^2+y^2} = \rho \sin \phi$$

$$\rho \cos \phi = \rho \sin \phi$$

$$\phi = \pi/4 \quad 0 \leq \phi \leq \pi/4$$

$$0 \leq \rho \leq 3\sqrt{2}$$

$$x^2+y^2=9$$



20 Use given transformations to evaluate the integral

a) $\iint_R (x-3y) dA$ where R is the triangular region with vertices $(0,0), (2,1)$

and $(1,2)$: $x = 2u+v$ $y = u+2v$

b) $\iint_R (4x+8y) dA$ where R is the parallelogram with vertices $(-1,3)$

$(1,-3)$ $(3,-1)$ and $(1,5)$

c) $\iint_R x^2 dA$ where R is the region bounded by the ellipse

$9x^2+4y^2=36$ $x=2u$ $y=3v$

d) $\iint_R (x^2 - xy + y^2) dA$ where R is the region bounded by the

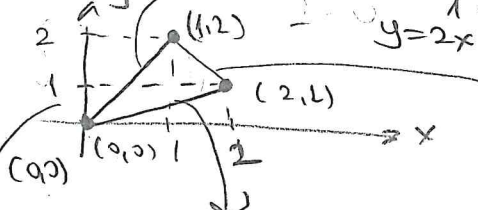
ellipse $x^2 - xy + y^2 = 2$; $x = \sqrt{2}u - \sqrt{2/3}v$, $y = \sqrt{2}u + \sqrt{2/3}v$

e) $J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 3$

$x-3y = 2u+v - 3(u+2v)$
 $= 2u+v - 3u-6v$
 $= -u-5v$

$|J| = |3| = 3$

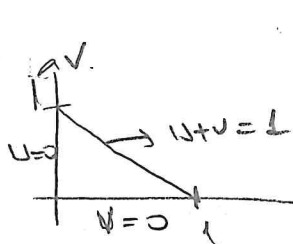
$y-0 = \frac{2-0}{1-0} (x-0) = 2x$



$y-2 = \frac{2-1}{1-2} (x-1) \Rightarrow y-2 = -(x-1)$
 $y = 2 - x + 1 = 3-x$

$y-0 = \frac{1-0}{2-0} (x-0) \Rightarrow y = \frac{x}{2}$

$y = 2x \Rightarrow u+2v = 2(2u+v)$
 $u+2v = 4u+2v$
 $u = 0$



$I = \int_0^1 \int_0^{1-u} (-5u-5v) |3| dv du$
 $I = -3 \int_0^1 \left[-5\frac{v^2}{2} - 5uv \right]_0^{1-u} du$

$y = \frac{x}{2} \Rightarrow u+2v = \frac{2u+v}{2}$

$2u+4v = 2u+v$
 $v = 0$

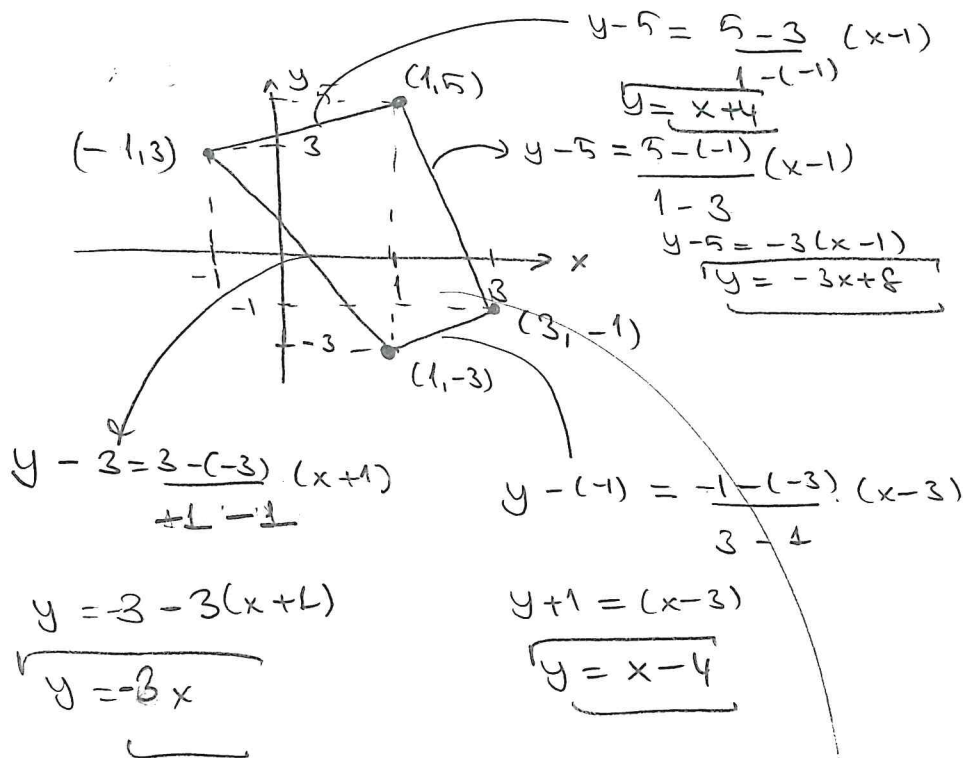
$I = -3 \int_0^1 \left[-5\frac{(1-u)^2}{2} - u(1-u) \right] du = 3-x \Rightarrow u+2v = 3 - (2u+v)$
 $u+2v = 3-2u-v$
 $3v+3u = 3$ $u+v=1$

$I = -3$

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(b) $x = \frac{1}{4}(u+v)$ $y = \frac{1}{4}(v-3u)$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 1/4 & 1/4 \\ -3/4 & 1/4 \end{vmatrix} = \frac{3}{16} + \frac{1}{16} = \frac{1}{4} \quad |J| = 1/4$$



$$\begin{cases} x = \frac{u}{4} + \frac{v}{4} \\ y = \frac{v}{4} - \frac{3u}{4} \end{cases}$$

$$y = x + 4$$

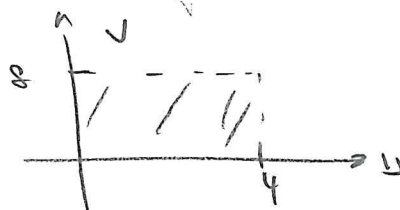
$$\frac{u}{4} + \frac{v}{4} = \frac{v}{4} - \frac{3u}{4} + 4$$

$$\frac{4u}{4} = 4 \quad \boxed{u=4}$$

$$y = -3x + 8$$

$$\frac{v}{4} - \frac{3v}{4} = -\frac{3v}{4} - \frac{3v}{4} + 8$$

$$\boxed{v=8}$$



$$\iint_R (4x + 8y) dA = \int_0^8 \int_0^4 (3v - 5u) \cdot \frac{1}{4} du dv = \int_0^8 \frac{1}{4} (3uv - \frac{5u^2}{2}) \Big|_0^4 dv$$

$$= 192.$$

$$4x + 8y = \frac{4u}{4} + \frac{4v}{4} + \frac{8v}{4} - \frac{24u}{4}$$

$$= u + v + 2v - 6u$$

$$= 3v - 5u$$

Deyan Altıntaş

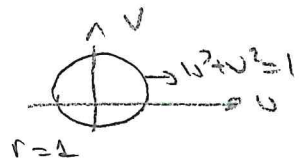
c) $\iint_D x^2 dA$ $D: 9x^2 + 4y^2 = 36$ $x = 2u$ $y = 3v$

$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 6$ $|J| = 6$

$9x^2 + 4y^2 = 36$ $\begin{matrix} x=2u \\ y=3v \end{matrix} \Rightarrow 36u^2 + 36v^2 = 36$ $u^2 + v^2 = 1$

$u = \cos \theta$

$v = \sin \theta$



2π

$\int_0^{2\pi} \int_0^1 64 \cdot \cos^2 \theta \cdot r^2 dr d\theta$

$x = 2u = 2 \cdot \cos \theta$

$x^2 = 4 \cos^2 \theta r^2$

$= \int_0^{2\pi} 64 \cdot \cos^2 \theta \cdot \frac{1}{4} d\theta = \frac{24}{4} \int_0^{2\pi} \cos^2 \theta d\theta$

$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$
 $= \cos^2 \theta - (1 - \cos^2 \theta)$

$= 6 \int_0^{2\pi} \frac{\cos 2\theta + 1}{2} d\theta = 6 \left(\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right) \Big|_0^{2\pi} = 6\pi$

d) $\iint_D (x^2 - xy + y^2) dA = ?$

$x^2 - xy + y^2 = 2$ $x = \sqrt{2} \left(u - \frac{v}{\sqrt{3}} \right)$

$y = \sqrt{2} \left(u + \frac{v}{\sqrt{3}} \right)$

$x^2 - xy + y^2 = 2 \Rightarrow 2 \left(u - \frac{v}{\sqrt{3}} \right)^2 - 2 \left(u^2 - \frac{v^2}{3} \right) + 2 \left(u + \frac{v}{\sqrt{3}} \right)^2 = 2$

$2 \left[u^2 - \frac{2uv}{\sqrt{3}} + \frac{v^2}{3} - u^2 + \frac{v^2}{3} + u^2 + \frac{2uv}{\sqrt{3}} + \frac{v^2}{3} \right] = 2(u^2 + v^2) = 2$

$u^2 + v^2 = 1$

$u = r \cos \theta$

$v = r \sin \theta$

$r = 1$

$x^2 - xy + y^2 = 2(u^2 + v^2)$

$u^2 + v^2 = r^2$

$\int_0^{2\pi} \int_0^1 \frac{2}{\sqrt{3}} r^2 \cdot r dr d\theta = \frac{4}{\sqrt{3}} \int_0^{2\pi} \frac{1}{4} d\theta = \frac{4}{\sqrt{3}} \cdot \frac{1}{2} \cdot 2\pi = \frac{\pi \sqrt{3}}{3}$

$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \sqrt{2} & -\frac{\sqrt{2}}{\sqrt{3}} \\ \sqrt{2} & \frac{\sqrt{2}}{\sqrt{3}} \end{vmatrix} = \frac{4}{\sqrt{3}}$

$|J| = \frac{4}{\sqrt{3}}$

